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MAT 3007 — Optimization

Solutions 8

Problem 1 (Non-Lischitz Smooth Function): (*20* points)
In Assignment 7, we have shown $f(x) = |x|^{3/2}$ is a convex function.

a) Prove f is not Lipschitz smooth. (Recall the definition: if $f : \mathbb{R}^n \to \mathbb{R}$ is Lipschitz smooth with constant $L > 0$, then $\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|$ for any $\boldsymbol{x}, \boldsymbol{y}$.)

b) Prove that for any constant step size $\bar{\alpha}$ in gradient descent method applied to this function, there exists x_0 for which $f(x_1) \geq f(x_0)$.

Remark: This problem highlights the importance of Lipschitz smoothness when using constant step size in the gradient descent method.

Solution :

a) If $x > 0$, then $f'(x) = \frac{3}{2}x^{1/2}$ and $f''(x) = \frac{3}{4}x^{-1/2} \to \infty$ as $x \to 0$. This can also be checked with definition.

b) We need to find x_0 such that

$$|x_1|^{3/2} = |x_0 - \bar{\alpha} f'(x_0)|^{3/2} > |x_0|^{3/2}.$$

We may assume that $x_0 > 0$. Then $f'(x_0) = \frac{3}{2}x_0^{1/2}$. We get

$$|x_1|^{3/2} = \left| x_0 - \frac{3}{2}\bar{\alpha}x_0^{1/2} \right|^{3/2} = |x_0|^{3/2} \left| \frac{3}{2}\bar{\alpha}x_0^{-1/2} - 1 \right|^{3/2}.$$

If $0 < x_0 < \frac{1}{4} \cdot \frac{9}{4}\bar{\alpha}^2 = \frac{9}{16}\bar{\alpha}^2$, then $\frac{3}{2}\bar{\alpha}x_0^{-1/2} - 1 > 1$ and $|x_1|^{3/2} > |x_0|^{3/2}$.

Problem 2 (Non-Strongly Convex Function): (*20* points)
Consider a function $f(x) = x^2 + 3\sin^2 x$.

a) Show that this function is not convex. Verify that $x^* = 0$ is a global minimizer.

b) Show that this function is Lipschitz smooth with some constant $L > 0$.

c) Consider the gradient descent method with a constant step size $\bar{\alpha}$. Show that after one iteration, the following inequality holds:

$$f(x_{k+1}) \leq f(x_k) - \left(\bar{\alpha} - \frac{L\bar{\alpha}^2}{2} \right) |f'(x_k)|^2.$$

You can directly use the following lemma.

Lemma 1 If g is a Lipschitz smooth function with constant L, then

$$g(\boldsymbol{y}) \leq g(\boldsymbol{x}) + \nabla g(\boldsymbol{x})^\top (\boldsymbol{y} - \boldsymbol{x}) + \frac{L}{2}\|\boldsymbol{y} - \boldsymbol{x}\|^2, \qquad (\star)$$

for any $\boldsymbol{x}$ and $\boldsymbol{y}$.

d) It can be shown that $|f'(x)|^2 \geq \frac{1}{16}(f(x) - f(x^*)), \forall x$ (you don't need to verify this). Combining this with part (c), show linear convergence in the following sense:

$$f(x_{k+1}) - f(x^*) \leq \eta[f(x_k) - f(x^*)].$$

for some $\eta \in (0, 1)$, with a proper step size $\bar{\alpha}$.

Remark: This problem shows that although the assumptions of Lipschitz smoothness and strong convexity are important for linear convergence, but they are not always necessary.

Solution :

a) Note that $f'(x) = 2x + 6\sin x \cos x$ and $f''(x) = 2 - 6\sin^2 x + 6\cos^2 x = 2 - 6\cos(2x)$. Since $f''(\pi/2) = -4$, the function is non-convex. Because $f(x) \geq 0$ for all x and $f(0) = 0$, then $x = 0$ is a global minimizer.

b) Note that $|f''(x)| = |2 - 6\cos(2x)| \leq 8$. Therefore, it is Lipschitz smooth with constant $L = 8$. This can also be shown using the definition.

c) Using the fact

$$f(x_{k+1}) \leq f(x_k) + f'(x_k)(x_{k+1} - x_k) + \frac{L}{2}|x_{k+1} - x_k|^2.$$

Plugging in $x_{k+1} - x_k = -\bar{\alpha} f'(x_k)$, we finish the proof.

d) From part (c),

$$f(x_{k+1}) - f(x^*) \leq f(x_k) - f(x^*) - \left(\bar{\alpha} - \frac{L\bar{\alpha}^2}{2}\right)|f'(x_k)|^2.$$

Then we plug in $|f'(x)|^2 \geq \frac{1}{16}(f(x) - f(x^*))$ and $L = 8$,

$$f(x_{k+1}) - f(x^*) \leq \left[1 - \frac{1}{16}\left(\bar{\alpha} - 4\bar{\alpha}^2\right)\right](f(x_k) - f(x^*)).$$

It has linear convergence as long as $\bar{\alpha} \in (0, 1/4)$. For example, if we take $\bar{\alpha} = 1/8$,

$$f(x_{k+1}) - f(x^*) \leq \frac{255}{256}(f(x_k) - f(x^*)).$$

Problem 3 (Bisection and Golden Section Methods): (*20* points)
Consider the optimization problem:

$$\min_x \; 4x^2 - x^4.$$

Starting from the interval $[-1, 1/2]$, solve the problem using both the Bisection Method and the Golden Section Method in MATLAB or Python. For both methods, output the midpoint of the final interval as the approximate solution. Compare the number of iterations required by each method to achieve a solution that is within 10^{-4} of the optimal solution in the given starting interval, $x^* = 0$.

Solution : After 15 iterations of the bisection method, it returns a solutions $x = -1.52587891 * 10^{-5}$ and the distance between left and right bounds is $9.16 * 10^{-5}$. After 20 iterations of the golden section method, it returns a solutions $x = 1.65267403 * 10^{-5}$ and the distance between left and right bounds is $9.92 * 10^{-5}$.

Problem 4 (True or False): (*20* points)
For each of the following statements, determine whether it is true or false. Provide a justification for your answer. Assume that we are minimizing a differentiable function f.

a) At the iterate $\boldsymbol{x}_k$, define a search direction $\boldsymbol{v}$ as follows:

$$\boldsymbol{v}_i = \begin{cases} -1 & \text{if the } i\text{-th coordinate of } \nabla f(\boldsymbol{x}_k) \text{ is positive,} \\ 1 & \text{if the } i\text{-th coordinate of } \nabla f(\boldsymbol{x}_k) \text{ is negative,} \\ 0 & \text{if the } i\text{-th coordinate of } \nabla f(\boldsymbol{x}_k) \text{ is zero.} \end{cases}$$

This search direction is a descent direction as long as $\boldsymbol{x}_k$ is not a stationary point.

b) In gradient descent with backtracking line search, it is always guaranteed that: $f(\boldsymbol{x}_{k+1}) < f(\boldsymbol{x}_k)$ as long as $\boldsymbol{x}_k$ is not a stationary point.

c) In gradient descent with exact line search, it is always guaranteed that: $f(\boldsymbol{x}_{k+1}) < f(\boldsymbol{x}_k)$ as long as $\boldsymbol{x}_k$ is not a stationary point.

d) Gradient descent will always converge to a local minimizer when using an exact line search.

Solution :

a) True. Note that $\boldsymbol{v}_i \nabla_i f(\boldsymbol{x}_k) < 0$ if $\nabla_i f(\boldsymbol{x}_k)$ is non-zero. Therefore, $\boldsymbol{v}^\top \nabla f(\boldsymbol{x}_k) < 0$ as long as $\nabla f(\boldsymbol{x}_k) \neq \mathbf{0}$.

b) True. Note that from Armijo condition:

$$f\left(\boldsymbol{x}_{k+1}\right) \leq f\left(\boldsymbol{x}_k\right) - \gamma\alpha \cdot \|\nabla f\left(\boldsymbol{x}_k\right)\|^2.$$

c) True. $-\nabla f(\boldsymbol{x}_k)$ is a descent direction if $\boldsymbol{x}_k$ is not a stationary point. Therefore, the function value decreases for sufficiently small step sizes. An exact line search determines the step size that achieves the greatest decrease in the function value along this direction.

d) False. Starting from a stationary point that is not a local minimizer, gradient descent will not move.

Problem 5 (Step Size Strategies): (*20* points)
Consider the optimization problem:

$$\min_x f(\boldsymbol{x}) = x^\top \boldsymbol{A} x, \quad \text{where } \boldsymbol{A} = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}.$$

Starting from the initial point $\boldsymbol{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, apply **gradient descent** using the update rule:

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \alpha_k \nabla f(\boldsymbol{x}_k)$$

with the following step size strategies:

a) Compute the following by hand or using a calculator:

- Perform two steps of gradient descent with a constant step size of 0.1, i.e., $\alpha_k = 0.1$ for all k.

- Perform one step of gradient descent with exact line search.
- Perform one step of gradient descent with backtracking line search: Search for α_k that satisfies the Armijo condition:

$$f\left(\boldsymbol{x}_k - \alpha \nabla f(\boldsymbol{x}_k)\right) \leq f\left(\boldsymbol{x}_k\right) - \gamma\alpha \cdot \|\nabla f\left(\boldsymbol{x}_k\right)\|^2$$

  If the condition is not satisfied, update $\alpha \leftarrow \sigma\alpha$. Use the parameters $\gamma = \frac{1}{2}$ and $\sigma = 0.2$.

b) Use Matlab or Python to implement the above three step size strategies and run gradient descent for 10 iterations.

Solution : Compute the gradient:

$$\nabla f(\boldsymbol{x}) = 2\boldsymbol{A}\boldsymbol{x} = \begin{bmatrix} 4x_1 \\ 10x_2 \end{bmatrix}.$$

a) **Constant step size.** Iteration 1:

$$\boldsymbol{x}_1 = \boldsymbol{x}_0 - \alpha_0 \nabla f(\boldsymbol{x}_0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.1 \times \begin{bmatrix} 4 \\ 10 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0 \end{bmatrix}.$$

Iteration 2:

$$\boldsymbol{x}_2 = \boldsymbol{x}_1 - \alpha_1 \nabla f(\boldsymbol{x}_1) = \begin{bmatrix} 0.6 \\ 0 \end{bmatrix} - 0.1 \times \begin{bmatrix} 2.4 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.36 \\ 0 \end{bmatrix}.$$

Exact line search. The step size is

$$\alpha_0 = \frac{\nabla f(\boldsymbol{x}_0)^\top \nabla f(\boldsymbol{x}_0)}{2\nabla f(\boldsymbol{x}_0)^\top \boldsymbol{A} \nabla f(\boldsymbol{x}_0)} = \frac{29}{266}.$$

This is because

$$\begin{aligned} \arg\min_{\alpha} f(\boldsymbol{x}_k - 2\alpha\boldsymbol{A}\boldsymbol{x}_k) &= \arg\min_{\alpha} \left(\boldsymbol{x}_k - 2\alpha\boldsymbol{A}\boldsymbol{x}_k\right)^\top \boldsymbol{A}(\boldsymbol{x}_k - 2\alpha\boldsymbol{A}\boldsymbol{x}_k) \\ &= \arg\min_{\alpha} \left[\boldsymbol{x}_k^\top \boldsymbol{A}\boldsymbol{x}_k + 4\alpha^2 \boldsymbol{x}_k^\top \boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{A}\boldsymbol{x}_k - 4\alpha \boldsymbol{x}_k^\top \boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{x}_k\right] \\ &= \frac{\boldsymbol{x}_k^\top \boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{x}_k}{2\boldsymbol{x}_k^\top \boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{A}\boldsymbol{x}_k} = \frac{\nabla f(\boldsymbol{x}_k)^\top \nabla f(\boldsymbol{x}_k)}{2\nabla f(\boldsymbol{x}_k)\boldsymbol{A}\nabla f(\boldsymbol{x}_k)} \end{aligned}$$

Then

$$\boldsymbol{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{29}{266} \times \begin{bmatrix} 4 \\ 10 \end{bmatrix} = \begin{bmatrix} 75/133 \\ -12/133 \end{bmatrix}.$$

Backtracking line search. Try $\alpha = 1, 0.2, 0.04, \ldots$. We find when $\alpha = 0.04$, the Armijo condition is satisfied:

$$\boldsymbol{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.04 \times \begin{bmatrix} 4 \\ 10 \end{bmatrix} = \begin{bmatrix} 0.84 \\ 0.6 \end{bmatrix},$$

and $f(\boldsymbol{x}_1) = 3.2112$. The Armijo condition:

$$f(\boldsymbol{x}_1) - f(\boldsymbol{x}_0) = -3.7888 \leq -0.5 \times 0.04 \times \left\| \begin{bmatrix} 4 \\ 10 \end{bmatrix} \right\|^2 = -2.32.$$

b) The results below are presented with six digits of precision. Using a constant step size, it outputs $\boldsymbol{x} = [0.006047, 0.000000]$ with $f(\boldsymbol{x}) = 0.000073$. Using exact line search, it outputs $\boldsymbol{x} = [0.000008, 0.000008]$, with $f(\boldsymbol{x}) = 0.000000$. For backtracking line search, the step size α_k can be initialized either from 1 or from the previous step size α_{k-1}. If the search restarts from 1, the result is $\boldsymbol{x} = [0.041643, -0.010078]$ with $f(\boldsymbol{x}) = 0.003976$. Without restarting the search (using α_{k-1} as the starting point), the result is $\boldsymbol{x} = [0.174901, 0.006047]$, with $f(\boldsymbol{x}) = 0.061364$.